DAY 20 · THEORY HANDOUT

Connect Python to a Local LLM

Use ChatOllama, invoke, streaming, and response metadata.

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The idea and its useful terms

Use ChatOllama, invoke, streaming, and response metadata.

LangChain

Python tools that connect prompts, models, and data.

ChatOllama

LangChain's chat wrapper for an Ollama server.

AIMessage

A reply object with text and other response details.

The wrapper gives us the same message interface for later chains. It sends the request to Ollama; it does not replace Ollama.

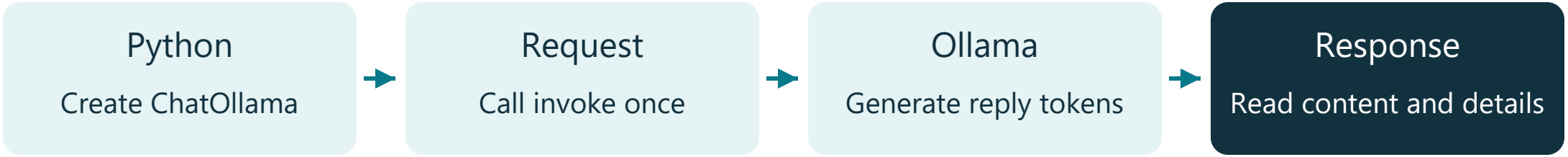
invoke returns one completed response. stream lets us read chunks as they arrive so a user can see progress.

> LangChain is a connection layer. The selected model still decides the reply quality.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

We create a ChatOllama object with a model tag and local server address. invoke sends our input and waits for an AIMessage. Reading response.content gives the reply text, separate from the other details.

stream returns chunks as generation proceeds. A chunk may contain little text or no text. Collect their content in order to rebuild the full reply. Streaming changes how we receive it, not the model's knowledge.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

Input

Explain a list in Python.

Keep the question short.

Response object

content holds reply text.

metadata holds run details.

Streaming

Chunks arrive over time.

Join them for full text.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> LangChain is a connection layer. The selected model still decides the reply quality.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

Model name

Must exist in Ollama.
Copy its exact tag.

base_url

Use localhost:11434
for the local server.

Environment

An environment variable
stores a setting outside code.

Import the wrapper, create a model object, then invoke one prompt.

Read content before looking at metadata. A token count is not a quality score.

> Stream a different prompt and join the chunks. Compare the two ways of receiving a reply.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Stream a different prompt and join the chunks. Compare the two ways of receiving a reply.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> LangChain is a connection layer. The selected model still decides the reply quality.

DAY 20 · COMPLETE

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